ECUAȚII DIFERENȚIALE

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I EC. DIF. CU VAR. SEPARABILE : $y' = h(x)g(y)$ y funcție de x !!

1. Rezolvăm $g(y)=0 \Rightarrow y = 0$ sol. const.
2. pt. $g(y) \neq 0$, Separăm după necunoscută:

$$\frac{y'(x)}{g(y(x))} = h(x) \quad / \int$$

$y'(x)\,dx \iff \frac{dy}{dx}dx = dy$

$$\int \frac{y'(x)\,dx}{g(y(x))} = \int h(x)\,dx$$

$G(y(x)) = H(x) + C$ sol. gen.

① $\frac{x}{y}\sqrt{1+y^2} + \sqrt{1+x^2}\cdot y' = 0$

$y' \cdot \sqrt{1+x^2} = -\frac{x}{y}\cdot\sqrt{1+y^2}$

$y' = -\frac{x}{y}\cdot\frac{\sqrt{1+y^2}}{\sqrt{1+x^2}}$

$y' = -\frac{x}{\sqrt{1+x^2}}\cdot\frac{\sqrt{1+y^2}}{y}$

$h(x)$ $g(y)$

1. $\frac{\sqrt{1+y^2}}{y} = 0$, $y \neq 0$

$\sqrt{1+y^2} = 0 \iff 1+y^2 = 0 \iff y = \pm i \notin \mathbb{R} \Rightarrow \nexists$ sol. const.

2. (y') $\frac{y}{\sqrt{1+y^2}} = -\frac{x}{\sqrt{1+x^2}} \quad / \int \qquad \int y' = \int dy$

$\int y'(x)\,dx = \int \frac{y}{\sqrt{1+y^2}}\,dy = -\int \frac{x}{\sqrt{1+x^2}}\,dx$

$= \int \frac{dy}{dx}\,dx = \int dy$

$\frac{1}{2}\int \frac{(1+y^2)'}{\sqrt{1+y^2}}\,dy = -\frac{1}{2}\int \frac{(1+x^2)'}{\sqrt{1+x^2}}\,dx \qquad \int \frac{dx}{\sqrt{x}} = 2\sqrt{x} + C$

$\sqrt{1+y^2} = -\sqrt{1+x^2} + c$, $c \in (0, +\infty)$ sol. gen.

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② $(1+x^2)y' = -xy$, $y(0)=2$

$y' = \frac{-xy}{(1+x^2)} = \underbrace{\frac{-x}{(1+x^2)}}_{h(x)} \cdot \underbrace{y}_{g(y)}$

1. $y=0$ sol. const.

2. $\frac{y'}{y} = \frac{-x}{(1+x^2)} \iff \frac{dy}{y} = \frac{-x}{(1+x^2)}dx \quad /\int$

$\int \frac{1}{y}dy = -\int \frac{x}{(1+x^2)}dx$

$\ln|y| = -\frac{1}{2}\ln(1+x^2) + c$

$\ln|y| = \ln(1+x^2)^{-\frac{1}{2}} + \ln e^c$

$\ln|y| = \ln \frac{e^c}{\sqrt{1+x^2}}$ mod. cu o altă const. K

$|y| = \frac{e^c}{\sqrt{1+x^2}} \Rightarrow y = \frac{\pm e^c}{\sqrt{1+x^2}} \to y = \frac{k}{\sqrt{1+x^2}}$

problemă Cauchy $y(0)=2 \Rightarrow y(0) = \frac{k}{\sqrt{1+0}} \Rightarrow k=2$

$y = \frac{2}{\sqrt{1+x^2}}$ sol. gen.

③ $y' = \frac{x^2 y}{1+y^2} \iff y' = \underbrace{x^2}_{h(x)} \cdot \underbrace{\frac{y}{1+y^2}}_{g(y)}$

1. $\frac{y}{1+y^2} = 0 \Rightarrow y=0$ sol. staționară

2. $\frac{y'(1+y^2)}{y} = x^2 \quad /\int$

$\int \frac{1+y^2}{y}dy = \int x^2 dx \iff \int \left(\frac{1}{y}+y\right)dy = \int x^2 dx \iff$

$\iff \ln|y| + y^2 \cdot \frac{1}{2} = \frac{x^3}{3} + c$ sol. gen.

④ $y' = \frac{x-xy}{x+1}$, $x > -1$

$y' = \underbrace{\frac{x(1-y)}{x+1}}_{h(x)} = \underbrace{\frac{x}{x+1}}_{g(y)} \cdot (1-y)$

1. $1-y=0 \to y=1$ sol. staționară

2. $\frac{y'}{(1-y)} = \frac{x}{x+1} \quad /\int \quad , y \neq -1$

$\int \frac{1}{1-y}dy = \int \frac{x}{x+1}dx$

$-\ln|1-y| = \int \frac{x+1-1}{x+1}dx = \int 1 - \frac{1}{x+1}dx = x - \ln(x+1) + c$

$\ln|1-y| = \ln(x+1) - x - c$

⑤ $(1+x^2)\cdot y \cdot y' + x(1+y^2) = 0$, $y(1)=2$

$y' = \underbrace{\frac{-x(1+y^2)}{y(1+x^2)}}_{h(x)} = \underbrace{\frac{-x}{1+x^2}}_{} + \underbrace{\frac{1+y^2}{y}}_{g(y)}$

1. $\frac{1+y^2}{y} = 0 \quad , y \neq 0 \Rightarrow 1+y^2 = 0 \Rightarrow y = \pm i \notin \mathbb{R} \Rightarrow \nexists$ sol. const.

2. $\frac{y'\cdot y}{1+y^2} = \frac{-x}{1+x^2} \quad /\int$

$\int \frac{y}{1+y^2}dy = -\int \frac{x}{1+x^2}dx$

$\frac{1}{2}\int \frac{2y}{1+y^2}dy = -\frac{1}{2}\int \frac{2x}{1+x^2}dx$

$\frac{1}{2}\ln(1+y^2) = -\frac{1}{2}\ln(1+x^2) + \ln e^c$

$\ln\sqrt{1+y^2} = \ln \frac{e^c}{\sqrt{1+x^2}}$

$\sqrt{1+y^2} = \frac{e^c}{\sqrt{1+x^2}} \quad /\uparrow^2$

$1+y^2 = \frac{e^{2c}}{1+x^2}$

$y = \pm\sqrt{\frac{e^{2c}}{1+x^2} - 1}$ sol. gen.

$\xrightarrow{y(1)=2} 2 = \pm\sqrt{\frac{k}{1+1} - 1} \Rightarrow \frac{k}{2} - 1 = 4 \Rightarrow k = 10$

$y = \pm\sqrt{\frac{10}{1+x^2} - 1}$

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⑥ $x^2(\sin y)\,y' + (\cos^2 y)\ln x = 0$, $\sin y \neq 0$, $x > 0$

$y' = \dfrac{-(\cos^2 y)\ln x}{x^2(\sin y)} = -\dfrac{\ln x}{x^2} \cdot \dfrac{\cos^2 y}{\sin y}$

1. $\dfrac{\cos^2 y}{\sin y} = 0 \Rightarrow \cos^2 y = 0 \Rightarrow y = \dfrac{\pi}{2} + k\pi$, $k \in \mathbb{R}$ sol. staționară

2. $\dfrac{y' \sin y}{\cos^2 y} = -\dfrac{\ln x}{x^2}$ $/\!/ \int$

$\int \dfrac{\sin y}{\cos^2 y}\,dy = -\int \dfrac{\ln x}{x^2}\,dx$

$\dfrac{1}{\cos y} + C_2 = \dfrac{\ln x + 1}{x} + C_1$

- $\int \dfrac{\ln x}{x^2}\,dx = -\dfrac{\ln x}{x} + \int \dfrac{1}{x^2} = -\dfrac{\ln x}{x} - \dfrac{1}{x} = -\dfrac{\ln x + 1}{x} + C_1$

$f = \ln x \Rightarrow f' = \frac{1}{x}$

$g' = \frac{1}{x^2} \Rightarrow g = -\frac{1}{x}$

- $\int \dfrac{\sin y}{\cos^2 y}\,dy = -\int \dfrac{(\cos y)'}{\cos^2 y}\,dy = \dfrac{1}{\cos y} + C_2$

$\dfrac{1}{\cos y} = \dfrac{1}{x}(\ln x + 1) + K$ sol. gen.

II. EC. DIF. OMOGENE : $y' = f\left(\frac{y}{x}\right)$

Se rezolvă folosind sch. de var. $z = \frac{y}{x}$

y, z funcții de x

- de obicei simplificăm forma inițială până ajungem la $\frac{y}{x}$
- după sch. de var. ec. devine o ec. dif. cu var. sep.

} $\Rightarrow$

① $xy' = y(\ln y - \ln x)$, $x, y > 0$

$y' = \dfrac{y}{x}\ln\dfrac{y}{x}$

not. $z = \dfrac{y}{x} \Rightarrow y = zx \Rightarrow y' = z'x + z$

$z'x + z = z\ln z$ a devenit ec. dif. cu var. sep.

$z' = z(\ln z - 1)\cdot\dfrac{1}{x}$

1. $z(\ln z - 1) = 0 \Rightarrow z = 0 \Rightarrow y = 0$ nu convine

$\ln z - 1 = 0 \Rightarrow z = e \Rightarrow y = ex$ sol. const.

2. $\dfrac{z'}{z(\ln z - 1)} = \dfrac{1}{x}$ $/\!/ \int$

$\int \dfrac{dz}{z(\ln z - 1)} = \int \dfrac{1}{x}\,dx$

$\ln|\ln z - 1| = \ln x + c$

$\ln\dfrac{\ln z - 1}{x} = c$

$\ln\dfrac{\ln\frac{y}{x} - 1}{x} = c$ sol. gen.

② $(x^2 + y^2)\,y' = xy$, $y(2) = 1$

$y' = \dfrac{xy}{(x^2+y^2)} \overset{(x^2}{=} \dfrac{\frac{y}{x}}{1 + (\frac{y}{x})^2}$

not. $\frac{y}{x} = z \Rightarrow y = xz \Rightarrow y' = z'x + z$

$z'x + z = \dfrac{z}{1+z^2}$

$z'x = \dfrac{z}{1+z^2} - z$

$z' = \dfrac{1}{x}\cdot\dfrac{z^3}{1+z^2}$ ec. dif. cu var. sep.

1. $\dfrac{z^3}{1+z^2} = 0 \Rightarrow z = 0 \Rightarrow y = 0$ sol. const.

2. $\dfrac{z'(1+z^2)}{z^3} = \dfrac{1}{x}$ $/\!/ \int$

$\int \dfrac{1+z^2}{z^3}\,dz = \int \dfrac{1}{x}\,dx$

$\int\left(\dfrac{1}{z^3} + \dfrac{1}{z}\right)dz = \ln|x| + c$

$-\dfrac{1}{z^2}\cdot\dfrac{1}{z^2} + \ln|z| = \ln|x| + c$

$-\dfrac{1}{2}\cdot\dfrac{x^2}{y^2} + \ln\left|\dfrac{y}{x}\right| = \ln|x| + c$

$\ln\left(e^{-\frac{x^2}{2y^2}}\cdot\dfrac{y}{x}\right) = \ln(x\cdot e^c)$

$e^{-\frac{x^2}{2y^2}}\cdot\dfrac{y}{x} = x\cdot K$

$y(2)=1 \Longrightarrow e^{-\frac{4}{2}}\cdot\dfrac{1}{2} = 2K \Rightarrow e^{-2} = 4K \Rightarrow K = \dfrac{1}{4e^2}$

$e^{-\frac{x^2}{2y^2}}\cdot\dfrac{1}{x} = x\cdot\dfrac{1}{4e^2}$

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③ $2x^2y' = y(2x^2 - y^2)$, $y(1)=1$, $x \neq 0$

$y' = \frac{y(2x^2-y^2)}{2x^3} \overset{(2x^2}{=} \frac{y}{x} \cdot \left(1 - \frac{1}{2}\left(\frac{y}{x}\right)^2\right) = \frac{1}{2} \cdot \frac{y}{x}\left(2 - \left(\frac{y}{x}\right)^2\right)$

not. $\frac{y}{x} = z \Rightarrow y = zx \Rightarrow y' = z'x + z$

$z'x + z = \frac{1}{2}z(2 - z^2)$

$z'x = -\frac{1}{2}z^3$

$z' = -\frac{z^3}{2x}$ ec. dif. cu var. sep.

1. $-z^3 = 0 \Rightarrow z = 0 \Rightarrow y = 0$ dar $y(1) = 0 \neq 1$ $\Rightarrow$ nu convine

2. $\frac{z'}{z^3} = -\frac{1}{2x} \quad / \int$

$\int \frac{1}{z^3}\,dz = -\frac{1}{2}\int \frac{1}{x}\,dx$

$\frac{z^{-2}}{-2} = -\frac{1}{2}\ln|x| + c$

$\frac{1}{2z^2} = \ln|x| + 2c$

$\frac{x^2}{y^2} = \ln|x| + k \overset{y(1)=1}{\Longrightarrow} 1 = \ln 1 + k \Leftrightarrow k = 1$

$y^2 = \frac{x^2}{\ln|x| + 1}$, $x \in (1, \infty)$

$y = \frac{x}{\sqrt{\ln x + 1}}$

④ $y' = \frac{4x - 3y}{3x - 2y} \overset{(x}{=} \frac{4 - 3\frac{y}{x}}{3 - 2\frac{y}{x}}$

not. $\frac{y}{x} = z \Rightarrow y = xz \Rightarrow y' = z'x + z$

$z'x + z = \frac{4 - 3z}{3 - 2z}$

$z' = \frac{4 - 3z - 3z + 2z^2}{3 - 2z} \cdot \frac{1}{x}$ ec. dif. cu var. sep.

1. $2z^2 - 6z + 4 = 0$, $z \neq \frac{3}{2}$

$\Delta = 36 - 32 = 4$

$z_1 = 2 \Rightarrow y_1 = 2x$

$z_2 = 1 \Rightarrow y_2 = x$ } sol. constr.

2. $\frac{z'(3-2z)}{(z-2)(z-1)} = \frac{1}{x} \quad / \int$

$\int \frac{3-2z}{(z-2)(z-1)}\,dz = \int \frac{1}{x}\,dx$

$\int \frac{3-2z}{2z^2 - 6z + 4}\,dz = \ln|x| + c$

$-\frac{1}{2}\int \frac{-6+4z}{2z^2-6z+4}\,dz = \ln|x| + c$

$-\frac{1}{2}\ln|2z^2 - 6z + 4| = \ln|x| + c$

$\ln \frac{1}{\sqrt{(z-2)(z-1)}} = \ln(x \cdot e^c)$

$K = x^2(z-2)(z-1)$

$K = x^2\left(\frac{y}{x} - 2\right)\left(\frac{y}{x} - 1\right)$

III EC. DIF. DE FORMA $y' = f(ax + by + c)$, $a, b, c \in \mathbb{R}$

Se rezolvă folosind sch. de var. $z = ax + by + c$ y, z f. de x

① $(x - y - 1) + y'(y - x + 2) = 0$

$y' = \frac{y - x + 1}{y - x + 2}$

$z = -x + y + 1 \Rightarrow y = x + z - 1 \Rightarrow y' = 1 + z'$

$1 + z' = \frac{z}{z+1}$

$z' = -1 + \frac{z}{z+1} = \frac{-z - 1 + z}{z+1} = \frac{-1}{z+1}$

$z' = \frac{-1}{z+1} \Leftrightarrow \frac{dz}{dx} = \frac{-1}{z+1} \Rightarrow (z+1)\,dz = -dx \quad / \int$

$\int (z+1)\,dz = -\int dx$

$\frac{1}{2}z^2 + z = -x + c$

$\frac{1}{2}(-x + y + 1)^2 + (-x + y + 1) = -x + c$

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② $(2x+y+1)\,dx - (4x+2y-3)\,dy = 0$

$y' = \frac{dy}{dx}$

$(2x+y+1)\,dx = (4x+2y-3)\,dy$

$\frac{(2x+y+1)}{(4x+2y-3)} \cdot \frac{dx}{dy} = 1$

$\frac{2x+y+1}{4x+2y-3} = y'$

not. $2x+y+1 = z \Rightarrow y = z-2x-1 \Rightarrow y' = z' - 2$

$z' - 2 = \frac{z}{2z-5}$

$z' = \frac{z+4z-10}{2z-5} = \frac{5z-10}{2z-5}$

$\frac{dz}{dx} = \frac{5z-10}{2z-5} \Rightarrow 5dx(z-2) = 2z-5\,dz \Rightarrow 5dx = \frac{2z-5}{z-2}\,dz \;/\int$

$5\int dx = \int \frac{2z-5}{z-2}\,dz = \int \left(\frac{2z-4}{z-2} - \frac{1}{z-2}\right) dz =$

$5x + 5C = 2z - \ln|z-2|$

... înlocuim $\Rightarrow$ sol. gen.

IV EC. DIF. DE FORMA $y' = \frac{a_1x+b_1y+c_1}{a_2x+b_2y+c_2}$

$\Delta \overset{not}{=} \begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix} \neq 0 \Rightarrow \exists!\,(x_0,y_0)$ sol. a.i. $\begin{cases} a_1x+b_1y+c_1=0 \\ a_2x+b_2y+c_2=0 \end{cases}$

Se rezolvă folosind sch. de var. $\boxed{z = \frac{y-y_0}{x-x_0}}$

① $y'(x+y-1)^2 = 2(y+2)^2$

$y' = \frac{2(y+2)^2}{(x+y-1)^2} \Rightarrow \Delta = \begin{vmatrix} 0 & 1 \\ 1 & 1 \end{vmatrix} = -1 \neq 0$

$\begin{cases} y+2=0 \\ x+y=1 \end{cases}$ (-)

$y+2-x-y = -1$

$x-2=1 \Rightarrow x=3 \Rightarrow y=-2 \Rightarrow (x_0,y_0) = (3,-2)$

Facem sch. de var. $z = \frac{y+2}{x-3} \Rightarrow y = z(x-3)-2$

$\Rightarrow y' = z'(x-3) + z$

$z'(x-3) + z = 2\frac{[z\cdot(x-3)]^2}{[z\cdot(x-3)-3+x]^2}$

$z'(x-3) = 2\frac{z^2(x-3)^2}{[z(x-3)+(x-3)]^2} - z$

$z'(x-3) = 2\frac{z^2(x-3)^2}{[(x-3)(z+1)]^2} - z$

$z'(x-3) = 2\frac{z^2}{(z+1)^2} - z$

$z'(x-3) = \frac{2z^2 - z(z+1)^2}{(z+1)^2}$

$z' = 2\frac{2z - z^2 - 2z - 1}{(z+1)^2(x-3)}$

$z' = z\frac{-(z^2+1)}{(z+1)^2(x-3)}$ ec. cu var. sep.

1. $\frac{-z(z^2+1)}{(z+1)^2} = 0 \;,\; z \neq -1$

$z=0 \Rightarrow z=0 \Rightarrow y=-2$ sol. singulară

~~$-(z^2+3z+1)=0$ nu e la fel pe ~~

2. $\frac{z'(z+1)^2}{-z(z^2+1)} = \frac{1}{x-3} \;/\int$

$-\int \frac{(z+1)^2}{z(z^2+1)}\,dz = \int \frac{1}{x-3}\,dx$

$-\int \left(\frac{1}{z} + \frac{2}{(z^2+1)}\right) dz = \ln|x-3| + C$

$-\ln|z| - 2\operatorname{arctg} z = \ln|x-3| + C$

$-\ln\left|\frac{y+2}{x-3}\right| - 2\operatorname{arctg}\left(\frac{y+2}{x-3}\right) = \ln|x-3| + C$ sol. gen.

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② $y' = \frac{x-y}{x+y+2}$

$\Delta = \begin{vmatrix} 0 & -1 \\ 1 & 1 \end{vmatrix} = 1 \neq 0$

$\begin{cases} x-y=0 \\ x+y+2=0 \end{cases} \Rightarrow (x_0, y_0) = (-1,-1)$

facem sch. de var. $z = \frac{y+1}{x+1} \Rightarrow y = z(x+1)-1 \Rightarrow y' = z'(x+1) + z$

$z'(x+1)+z = \frac{-z(x+1)+1+x}{z(x+1)+1+x} = \frac{(x+1)(1-z)}{(x+1)(z+1)} = \frac{1-z}{1+z}$

$z'(x+1) = \frac{1-2z-z^2}{1+z}$

$z' = \frac{-z^2-2z+1}{(1+z)(x+1)}$ ec. dif cu var. sep.

1. $-(z^2+2z-1) = 0$

$z = -1 \pm \sqrt{2} \Rightarrow y = \dots$

2. $\frac{z'(1+z)}{-(z^2+2z-1)} = \frac{1}{x+1} \;/\int \dots$

V EC. DIF. LINIARE DE ORD. I : $y' + f(x)\cdot y = g(x)$

1. Se înmulțește cu $e^{\int f(x)\,dx}$

$\Rightarrow y' e^{\int f(x)dx} + f(x)\cdot y \cdot e^{\int f(x)dx} = g(x)\cdot e^{\int f(x)dx}$

$\left(y \cdot e^{\int f(x)dx}\right)' = g(x)\cdot e^{\int f(x)dx}$

2. Se rezolvă mai ușor ec:

$y \cdot e^{\int f(x)dx} = \int \left(g(x)\cdot e^{\int f(x)dx}\right) dx$

$$y = \frac{\int \left(g(x)\cdot e^{\int f(x)dx}\right) dx}{e^{\int f(x)dx}}$$

① $x(x-1)y' + 2xy = 1, \quad x>1$

$y' + \underbrace{\frac{2y}{x-1}}_{f(x)} = \underbrace{\frac{1}{x(x-1)}}_{g(x)} \;/\; e^{\int \frac{2}{x-1}dx} = e^{2\ln(x-1)} = (x-1)^2$

$y'(x-1)^2 - 2y(x-1) = \frac{x-1}{x}$

$\left(y(x-1)^2\right)' = \frac{x-1}{x} \;/\int$

$y(x-1)^2 = \int \frac{x-1}{x}dx = \int \left(1-\frac{1}{x}\right)dx = x - \ln x + c$

$y = \frac{x-\ln x + c}{(x-1)^2}$ sol. gen.

SAU cu formula

$y = \frac{\int \left(\frac{1}{x(x-1)}\cdot e^{\int \frac{2}{x-1}dx}\right)dx}{e^{\int \frac{2}{x-1}dx}} = \frac{\int \frac{1}{x(x-1)}(x-1)^2\,dx}{(x-1)^2} = \frac{x-\ln x+c}{(x-1)^2}$

② $(xy+e^x)\,dx - x\,dy = 0$

$y' = \frac{dy}{dx}$

$(xy+e^x)\,dx = x\,dy \Rightarrow \frac{xy+e^x\,dx}{x\,dy} - 1 \Rightarrow y' = \frac{xy+e^x}{x}$

$y' = y + \underbrace{\frac{e^x}{x}}_{g(x)} \quad \underbrace{f(x)=-1}\; / \; e^{-\int dx} = e^{-x}$

$y'e^{-x} - ye^{-x} = \frac{1}{x}$

$(ye^{-x})' = \frac{1}{x} \;/\int$

$ye^{-x} = \ln|x| + c$

$y = \frac{\ln|x|+c}{e^{-x}}$ sol. gen.

③ $y' = \frac{y}{2x} + e^{3x}\cdot\sqrt{x}, \quad x>0$

$y' \ominus y\cdot\underbrace{\left(\frac{1}{2x}\right)}_{f(x)} = \underbrace{e^{3x}\sqrt{x}}_{g(x)} \;/\; e^{-\frac{1}{2}\int \frac{1}{x}dx} = e^{-\frac{1}{2}\ln x} = e^{\ln \frac{1}{\sqrt{x}}} = \frac{1}{\sqrt{x}}$

$y'\frac{1}{\sqrt{x}} - \frac{y}{2x\sqrt{x}} = e^{3x}$

$\left(y\frac{1}{\sqrt{x}}\right)' = e^{3x} \;/\int$

$y\frac{1}{\sqrt{x}} = \frac{1}{3}e^{3x} + c$ sol. gen.

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④ $y' + y\underbrace{\sin x}_{f(x)} = \underbrace{-\sin x \cos x}_{g(x)} \quad / \cdot e^{\int \sin x\,dx} = e^{-\cos x}$

$y' e^{-\cos x} + y \sin x\, e^{-\cos x} = -\sin x \cos x \cdot e^{-\cos x}$

$(y e^{-\cos x})' = -\sin x \cos x \cdot e^{-\cos x} \quad / \int$

$y e^{-\cos x} = -\int \sin x \cos x \cdot e^{-\cos x}\,dx =$

$= +\int (\cos x)' \cos x \cdot e^{-\cos x}\,dx \overset{\cos x = t}{=}$

$= \int t e^{-t}\,dt =$

$= -te^{-t} + \int e^{-t} =$

$= -te^{-t} - e^{-t} =$

$= -e^{-t}(t+1)$

$y e^{-\cos x} = -e^{-\cos x}(\cos x + 1) + c$

$y = -1 - \cos x + c e^{\cos x}$ sol. gen.

VI EC. DIF. BERNOULLI : $y' = f(x) \cdot y + g(x) \cdot y^\alpha$, $\alpha \in \mathbb{R}$

1. Înmulțim ec. cu $\frac{1}{y^\alpha}$

$\frac{1}{y^\alpha} y' = \frac{1}{y^\alpha} f(x) y + \frac{1}{y^\alpha} g(x) y^\alpha$

$y' y^{-\alpha} = f(x) y^{1-\alpha} + g(x)$

2. Înmulțim ec. cu $1-\alpha$

$(1-\alpha) y^{-\alpha} y' = (1-\alpha) y^{1-\alpha} f(x) + (1-\alpha) g(x)$

3. Facem sch. de var. $z = y^{1-\alpha} \Rightarrow z' = (1-\alpha) y^{-\alpha} \cdot y'$

$z' = (1-\alpha) f(x) z + (1-\alpha) g(x)$ ec. dif. lin. de ord. I

① $y' + \underbrace{x}_{f(x)} y = \underbrace{e^{-2x^2}}_{g(x)} y^3$, $y \neq 0 \quad / \cdot \frac{1}{y^{-3}} = y^3$

$y' y^3 + x y^4 = e^{-2x^2} \quad / \cdot (1+3)$

$4y^3 y' + 4y^4 x = 4e^{-2x^2}$

Facem sch. de var. $z = y^4 \Rightarrow z' = 4y^3 \cdot y'$

$z' + 4xz = 4e^{-2x^2}$ ec. lin. ord. I

$z' + \underbrace{4x}_{f(x)} z = \underbrace{4e^{-2x^2}}_{g(x)} \quad / \cdot e^{\int 4x\,dx} = e^{4\frac{x^2}{2}} = e^{2x^2}$

$z' e^{2x^2} + 4xz e^{2x^2} = 4e^{-2x^2} e^{2x^2}$

$(z e^{2x^2})' = 4 \quad / \int$

$z e^{2x^2} = 4x + c$

$z = \frac{4x + c}{e^{2x^2}}$

$y = \sqrt[4]{\frac{4x + c}{e^{2x^2}}}$ sol. gen.

② $2y' - \frac{x}{y} = \frac{xy}{x^2-1}$

$y' - \frac{x}{2y} = \frac{xy}{2(x^2-1)}$

$y' = \underbrace{\frac{x}{2(x^2-1)}}_{f(x)} y + \underbrace{\frac{x}{2}}_{g(x)} \cdot y^{-1} \quad \alpha = -1 \quad / \cdot \frac{1}{y^{-1}} = y$

$yy' = \frac{x}{2(x^2-1)} y^2 + \frac{x}{2} \quad / \cdot (1+1)$

$2yy' = \frac{x}{x^2-1} y^2 + x$

Facem sch. de fct. $z = y^2 \Rightarrow z' = 2y \cdot y'$

$z' = \frac{x}{x^2-1} z + x$ ec. dif. lin. ord. I

$z' \ominus \underbrace{\frac{x}{x^2-1}}_{f(x)} z = \underbrace{x}_{g(x)} \quad / \cdot e^{-\int \frac{x}{x^2-1}\,dx} = e^{-\frac{1}{2}\int \frac{2x}{x^2-1}\,dx} = e^{-\frac{1}{2}\ln|x^2-1|}$

$= e^{\ln \frac{1}{\sqrt{x^2-1}}} = \frac{1}{\sqrt{x^2-1}}$

$\frac{z'}{\sqrt{x^2-1}} - \frac{x}{(x^2-1)\sqrt{x^2-1}} z = \frac{x}{\sqrt{x^2-1}}$

$\left(\frac{z}{\sqrt{x^2-1}}\right)' = \frac{x}{\sqrt{x^2-1}} \quad / \int$

$\frac{z}{\sqrt{x^2-1}} = \int \frac{x}{\sqrt{x^2-1}}\,dx = \frac{1}{2}\int \frac{2x}{\sqrt{x^2-1}}\,dx = \frac{1}{2}\sqrt{x^2-1}\Big|_{\frac{1}{2}} + c$

$z = x^2 - 1 + c\sqrt{x^2-1}$ sol. gen.

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③ $y' = \frac{y}{x} + \frac{x^3}{3y^2}$, $x, y \neq 0$, pres. $x > 0$

$y' = y \cdot \frac{1}{x} + \frac{x^3}{3} \cdot y^{-2}$ $\alpha = -2$ $\quad / \cdot \frac{1}{y^{-2}} = y^2$

$f(x)$ $g(x)$

$y^2 y' = y^3 \frac{1}{x} + \frac{x^3}{3}$ $\quad / \cdot (1+2)$

$3y^2 y' = 3y^3 \frac{1}{x} + x^3$

facem sch. de fct.: $z = y^3 \Rightarrow z' = 3y^2 \cdot y'$

$z' = 3z \frac{1}{x} + x^3$

$z' - 3z \left(\frac{1}{x}\right) = x^3$ ec. dif. lim. ord. I $\quad / \cdot e^{-3\int \frac{1}{x}} = e^{-3\ln x} = e^{\ln \frac{1}{x^3}} = \frac{1}{x^3}$

$f(x)$ $g(x)$

$z' \cdot \frac{1}{x^3} - 3z \cdot \frac{1}{x^4} = 1$

$\left(z \cdot \frac{1}{x^3}\right)' = 1$ $\quad / \int$

$z \cdot \frac{1}{x^3} = x + c$

$z = x^4 + cx^3$

$y = \sqrt[3]{x^4 + cx^3} = x\sqrt[3]{x+c}$, $c \in \mathbb{R}$

④ $y' = y^4 \cos x + y \operatorname{tg} x$, $x \in (0, \frac{\pi}{2})$ $\quad / \cdot \frac{1}{y^4}$ $\quad \alpha = 4$

$g(x)$ $f(x)$

$\frac{y'}{y^4} = \frac{\operatorname{tg} x}{y^3} + \cos x$ $\quad / \cdot (1-4) = -3$

$\frac{-3y'}{y^4} = \frac{-3 \operatorname{tg} x}{y^3} - 3\cos x$

facem sch. de var. $z = y^{-3} \Rightarrow z' = -3y^{-4} \cdot y'$

$z' = \frac{-3 \operatorname{tg} x}{z^{-1}} - 3\cos x$

$z' + z\, 3\operatorname{tg} x = -3\cos x$ ec. dif. lim. ord. I $\quad / e^{3\int \operatorname{tg} x\, dx} =$

$f(x)$ $g(x)$

$= e^{-3\ln(\cos x)} = \frac{1}{\cos^3 x}$

$\frac{z'}{\cos^3 x} + \frac{z\, 3\operatorname{tg} x}{\cos^3 x} = \frac{-3}{\cos^2 x}$

$\left(\frac{z}{\cos^3 x}\right)' = \frac{-3}{\cos^2 x}$ $\quad / \int$

$(\operatorname{tg} x)' = \frac{1}{\cos^2 x}$!

$\frac{z}{\cos^3 x} = -3 \int \frac{1}{\cos^2 x} dx = -3 \operatorname{tg} x + c \Rightarrow y^{-3} = -3 \operatorname{tg} x \cos^3 x + c\cos^3 x$

EC. DIF. DE ORDIN SUPERIOR cu coef. const.

I ec. este egală cu un **QUASIPOLINOM** (număr de cvasipolinoame)

$e^{\alpha x}[P_m(x)\cos(\beta x) + Q_m(x)\sin(\beta x)]$

forma soluției va fi: $y = y_h(x) + y_p(x)$, unde:

→ y_h este sol. gen. a ec. dif. omogene obținută

→ y_p este sol. particulară a ec. dif. neomogene

y_h:

1. Se scrie ec. caract. omogenă obținută în r și i se det. sol. (ec. va avea r în loc de y, iar puterea lui r va fi ordinul derivatei lui)
2. Se scrie m_r pentru fiecare r
3. Se scrie o bază pt. spațiul sol. ec. dif. omogene, de forma

SE FACE PT. FIECARE r: $\{y_1 = e^{rx}, y_2 = xe^{rx}, y_3 = x^2 e^{rx}, \dots, y_m = x^{m-1} e^{rx}\}$ pt. o sol. r cu $m_r = m$!

4. Sol. gen. y_h va fi $y_h = c_1 y_1(x) + c_2 y_2(x) + \dots + c_m y_m(x)$; $c_1, \dots, c_m \in \mathbb{R}$

y_p:

1. Se egalează funcția de după egal cu un cvasipolinom și se identifică α, β, m, n
2. Se identifică $M = \max\{m, n\}$
3. Se va căuta sol. particulară y_p a ec. dif. neomogene de forma

$e^{\alpha x} \cdot x^k [\hat{P}_M(x)\cos(\beta x) + \hat{Q}_M(x)\sin(\beta x)]$

4. k este multiplicitatea sol. $\alpha + i\beta$ în ec. caract. dacă $\alpha + i\beta \neq$ sol $\Rightarrow k = 0$

(*) dacă ec. este egală cu sumă de cvasipolinoame, se separă cele 2 ec. și se află y_p pt. fiecare: $y = y_h(x) + y_{p_1}(x) + y_{p_2}(x) + \dots$

II ec. **NU** este egală cu un **CVASIPOLINOM**

y_h: ! y_h se abordează identic cu I

y_p: **METODA VARIAȚIEI CONSTANTELOR**

1. y_p va proveni din y_h, fiecare constantă $c_1, \dots, c_m$ devenind funcție de x („vorbim constantelor")
2. deci căutăm $y_p = c_1(x) y_1(x) + c_2(x) y_2(x) + \dots + c_m(x) y_m(x)$

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3. Se rezolvă sistemul:

$$\begin{cases} c_1'y_1 + c_2'y_2 + \dots + c_m'y_m = 0 \\ c_1'y_1' + c_2'y_2' + \dots + c_m'y_m' = 0 \\ \vdots \\ c_1'y_1^{(m-1)} + c_2'y_2^{(m-1)} + \dots + c_m'y_m^{(m-1)} = \frac{f(x)}{a_m} \end{cases}$$

funcția de după egal

coef. lui y derivată de cele mai multe ori

① $y'' - 5y' = \underbrace{3x^2 + \sin 5x}_{\text{sumă de cvasipolinoame}}$

• y_h

$r^2 - 5r = 0$

$r(r-5) = 0$

$r_1 = 0 \Rightarrow m_{r_1} = 1$

$r_2 = 5 \Rightarrow m_{r_2} = 1$

$y_1(x) = 1, \ y_2(x) = e^{5x}$

$y_h = c_1 + c_2 e^{5x}$

• y_{p_1}

$y'' - 5y' = 3x^2$

$f(x) = 3x^2 = e^{\alpha x}[P_m(x)\cos\beta x + Q_m(x)\sin\beta x]$

$\alpha = 0, \beta = 0 \Rightarrow \alpha + i\beta = 0$ este rad. a ec. caract. $\Rightarrow l = 1$

$P_m(x) = 3x^2 \Rightarrow m = 2$

$Q_m(x) = 0 \Rightarrow m = -\infty$ $\Big\} \ l = \max\{m, m\} = 2$

$\Rightarrow y_{p_1} = x(a + bx + cx^2) = ax + bx^2 + cx^3$

$y_{p_1}' = a + 2bx + 3cx^2$

$y_{p_1}'' = 2b + 6cx$

$y_{p_1}'' - 5y_{p_1}' = 3x^2 \Rightarrow 2b + 6cx - 5a - 10bx - 15cx^2 = 3x^2$

$$\begin{cases} -15c = 3 \Rightarrow c = -\frac{1}{5} \\ 6c - 10b = 0 \Rightarrow b = -\frac{3}{25} \\ 2b - 5a = 0 \Rightarrow a = -\frac{6}{125} \end{cases}$$

$y_{p_1} = -\frac{6}{125}x - \frac{3}{25}x^2 - \frac{1}{5}x^3$

• y_{p_2}

$y'' - 5y' = \sin 5x$

$f(x) = \sin 5x = e^{\alpha x}[P_m(x)\cos\beta x + Q_m(x)\sin\beta x]$

$\alpha = 0, \beta = 5 \Rightarrow \alpha + i\beta = 5i$ nu este rad. a ec. caract.

$P_m(x) = 0 \Rightarrow m = -\infty$

$Q_m(x) = 1 \Rightarrow m = 0$ $\Big\} \Rightarrow l = \max\{m, m\} = 0$

$\Rightarrow y_{p_2} = a\cos 5x + b\sin 5x$

$y_{p_2}' = -5a\sin 5x + 5b\cos 5x$

$y_{p_2}'' = -25a\cos 5x - 25b\sin 5x$

$y'' - 5y' = \sin 5x \Rightarrow -25a\cos 5x - 25b\sin 5x + 25a\sin 5x - 25b\cos 5x = \sin 5x$

$$\begin{cases} +25a - 25b = 1 \Rightarrow a = \frac{1}{25} + b \\ -25a - 25b = 0 \Rightarrow a = -b \end{cases} \Rightarrow -b = \frac{1}{25} + b \Rightarrow b = \frac{1}{50}, \quad a = -\frac{1}{50}$$

$y_{p_2} = -\frac{1}{50}\cos 5x + \frac{1}{50}\sin 5x$

$y = y_h + y_{p_1} + y_{p_2} = c_1 + c_2 e^{5x} - \frac{6}{125}x - \frac{3}{25}x^2 - \frac{1}{5}x^3 + \frac{1}{50}(\cos 5x + \sin 5x)$

② $y^{(5)} - y^{(4)} + y^{(3)} = 0$

ec. caract.:

$r^5 - r^4 + r^3 = 0$

$r^3(r^2 - r + 1) = 0$

$r_1 = 0, \ m_{r_1} = 3 \Rightarrow \{e^{0x}, xe^{0x}, x^2e^{0x}\} \Leftrightarrow \{1, x, x^2\}$

$r^2 - r + 1 = 0$

$\Delta = 1 - 4 = -3 \Rightarrow \sqrt{\Delta} = i\sqrt{3}$

$r_{2/3} = \frac{1 \pm i\sqrt{3}}{2} \Rightarrow r_2 = \frac{1 + i\sqrt{3}}{2}$ ($\alpha = \frac{1}{2}$, $\beta = \frac{\sqrt{3}}{2}$), $m_{r_2} = 2 \Rightarrow \left\{e^{\frac{1}{2}x}\cos\left(\frac{\sqrt{3}}{2}x\right), e^{\frac{1}{2}x}\sin\left(\frac{\sqrt{3}}{2}x\right)\right.$

$y(x) = c_1 + c_2 x + c_3 x^2 + c_4 e^{\frac{1}{2}x}\cos\frac{\sqrt{3}}{2}x + c_5 e^{\frac{1}{2}x}\sin\frac{\sqrt{3}}{2}x$

• p.t. $r \in \mathbb{C} = \alpha + i\beta$ cu $m_r = m$, avem baza

$\{y_1 = e^{\alpha x}\cos\beta x, \ y_2 = e^{\alpha x}\sin\beta x, \ y_3 = xe^{\alpha x}\cos\beta x, \ y_4 = xe^{\alpha x}\sin\beta x\}$

• dacă $r \in \mathbb{C}$ apare sub forma $r = \alpha \pm i\beta$, considerăm doar $r = \alpha + i\beta$ cu $m_r = 2$ și formăm baza

3.02.2025

③ $y'' - 9y = e^{3x}\cos x$

• y_h

rezolv ec. caract.:

$r^2 - 9 = 0$

$r = \pm 3$

$r_1 = 3,\ m_{r_1} = 1$

$r_2 = -3,\ m_{r_2} = 1$

e^{3x}, e^{-3x} sol. fundamentale

$\{y_1 = e^{3x}, y_2 = e^{-3x}\}$ o bază pt. spațiul sol. ec. dif. omogene

$y_h = c_1 e^{3x} + c_2 e^{-3x}$, $c_1, c_2 \in \mathbb{R}$

• y_p

$f(x) = e^{3x}\cos x = e^{\alpha x}[P_m(x)\cos\beta x + Q_m(x)\sin\beta x]$

$\alpha = 3,\ \beta = 1 \Rightarrow \alpha + i\beta = 3+i$ nu este sol. a ec. caract. $\Rightarrow s = 0$

$P_m(x) = 1 \Rightarrow m = 0$

$Q_m(x) = 0 \Rightarrow m = -\infty$ $\Rightarrow M = \max\{m, m\} = 0$

$y_p = x^s e^{\alpha x}[\tilde{P}_M(x)\cos\beta x + \tilde{Q}_M(x)\sin\beta x]$

$y_p = e^{3x}[\tilde{P}_0(x)\cos\beta x + \tilde{Q}_0(x)\sin\beta x]$

$y_p = e^{3x}(a\cos\beta x + b\sin\beta x)$

$y_p' = 3e^{3x}(a\cos\beta x + b\sin\beta x) + e^{3x}(-a\sin\beta x + b\cos\beta x)$

$y_p'' = 9e^{3x}(a\cos\beta x + b\sin\beta x) + 3e^{3x}(-a\sin\beta x + b\cos\beta x) + 3e^{3x}(-a\sin\beta x + b\cos\beta x) + e^{3x}(-a\cos\beta x - b\sin\beta x)$

$y'' - 9y = e^{3x}\cos x$

$6e^{3x}(-a\sin\beta x + b\cos\beta x) + e^{3x}(-a\cos\beta x - b\sin\beta x) = e^{3x}\cos x$

$\begin{cases} -6a - b = 0 \Rightarrow b = -6a \\ 6b - a = 1 \end{cases}$ $\Rightarrow -36a - a = 1 \Rightarrow a = -\frac{1}{37}$

$\beta = \frac{6}{37}$

$y_p = e^{3x}\left(-\frac{1}{37}\cos\beta x + \frac{6}{37}\sin\beta x\right)$

$y(x) = y_h(x) + y_p(x) = \ldots$

④ $y^{(4)} - 2y^{(3)} + y'' = e^x$

y_h

rezolv. ec. caract.

$r^4 - 2r^3 + r^2 = 0$

$r^2(r^2 - 2r + 1) = 0$

$r_1 = 0,\ m_{r_1} = 2 \Rightarrow e^{0x}, xe^{0x}$

$r_2 = 1,\ m_{r_2} = 2 \Rightarrow e^{1x}, xe^{1x}$ } sol. fundamentale

$\Rightarrow$ o bază pt. spațiul soluțiilor ec. dif. omogene asociate este

$\{y_1 = 1, y_2 = x, y_3 = e^x, y_4 = xe^x\}$

$y_h(x) = c_1 + c_2 x + c_3 e^x + c_4 x e^x$, $c_1, \ldots, c_4 \in \mathbb{R}$

y_p

$f(x) = e^x = e^{\alpha x}[P_m(x)\cos\beta x + Q_m(x)\sin\beta x]$

$\alpha = 1,\ \beta = 0 \Rightarrow \alpha + i\beta = 1$ este sol. a ec. caract. $\Rightarrow s = 2$

$P_m(x) = 1 \Rightarrow m = 0$

$Q_m(x) = 0 \Rightarrow m = -\infty$ $\Rightarrow M = \max\{m, m\} = 0$

$y_p = x^s e^{\alpha x}[\tilde{P}_M(x)\cos\beta x + \tilde{Q}_m(x)\sin\beta x]$

$y_p = x^2 e^x \cdot a$

$y_p^{(4)} - 2y_p^{(3)} + y_p'' = e^x$

$y_p' = ae^x \cdot 2x + x^2 \cdot ae^x = ae^x(x^2 + 2x)$

$y_p'' = ae^x \cdot 2 + 2xae^x + 2xae^x + ae^x x^2 = ae^x(x^2 + 4x + 2)$

$y_p^{(3)} = 2ae^x + 2ae^x + 2xae^x + 2ae^x + 2xae^x + 2xae^x + x^2 ae^x =$

$= 6ae^x + 6xae^x + ax^2 e^x = ae^x(x^2 + 6x + 6)$

~~$\Rightarrow ae^x(x^2 + 6x + 6) - 2ae^x(x^2 + 4x + 2) + ae^x(x^2 + 2x) = e^x$~~

$y_p^{(4)} = ae^x(x^2 + 6x + 6) + ae^x(2x + 6) = ae^x(x^2 + 8x + 12)$

$ae^x(x^2 + 8x + 12) - 2ae^x(x^2 + 6x + 6) + ae^x(x^2 + 4x + 2) = e^x$

$ae^x(x^2 + 8x + 12 - 2x^2 - 12x - 12 + x^2 + 4x + 2) = e^x$

$2ae^x = e^x$

$a = \frac{1}{2}$

$y_p = \frac{1}{2}x^2 e^x$

$y(x) = y_h(x) + y_p(x) = \ldots$

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⑤ $y'' + 2y' + y = xe^x + \frac{1}{xe^x}$

• y_h

rezolv. ec. caract.

$r^2 + 2r + 1 = 0$

$(r+1)^2 = 0$

$r = -1$, $m_r = 2$ $\Rightarrow$ e^{-x}, xe^{-x} sol. fundamentale

$\{f_1 = e^{-x}, f_2 = xe^{-x}\}$ o bază pt. spațiul sol. ec. dif. omogene asociată

$y_h = c_1 e^{-x} + c_2 xe^{-x}$, $c_1, c_2 \in \mathbb{R}$

• y_{p_1}

$f_1(x) = \frac{1}{xe^x}$ nu este cvasipolinom

folosim metoda variației constantelor

$y_{p_1} = c_1(x) e^{-x} + c_2(x) xe^{-x}$

rezolvăm sistemul:

$$\begin{cases} c_1' e^{-x} + c_2' xe^{-x} = 0 \\ c_1'(-e^{-x}) + c_2'(e^{-x} - e^{-x}x) = \frac{1}{xe^x} \end{cases}$$

$$\begin{cases} c_1' e^{-x} + c_2' xe^{-x} = 0 \\ -c_1' e^{-x} + c_2' e^{-x}(1-x) = \frac{1}{xe^x} \end{cases}$$

(+)

$c_2' e^{-x}(x + 1 - x) = \frac{1}{xe^x}$

$c_2' = \frac{1}{e^{-x} xe^x} = \frac{1}{x}$ / $\int$

$c_2 = \ln|x| + k_1$

$c_1' + \frac{1}{x} x = 0 \Rightarrow c_1' = -1$ / $\int \Rightarrow c_1 = -x + k_2$ $\xrightarrow{k_1 + k_2 = 0}$

$y_{p_1} = -xe^{-x} + \ln|x| xe^{-x}$

• y_{p_2} cvasipolinom

$f_2(x) = xe^x = e^{\alpha x}[P_m(x)\cos\beta x + Q_m(x)\sin\beta x]$

$\alpha = 1$, $\beta = 0 \Rightarrow \alpha + i\beta = 1$ nu este rădăc. a ec. caract. $\Rightarrow s = 0$

$P_m(x) = x \Rightarrow m = 1$

$m(x) = 0 \Rightarrow m = -\infty$ $\Rightarrow M = \max\{m, m\} = 1$

$y_{p_2} = x^s e^{\alpha x}[\tilde{P}_M(x)\cos\beta x + \tilde{Q}_M(x)\sin\beta x]$

$y_{p_2} = e^x(ax + b)$

$y_{p_2}' = e^x(ax + b) + e^x(a) = e^x(ax + b + a)$

$y_{p_2}'' = e^x(ax + b + a) + e^x(a) = e^x(ax + b + 2a)$

~~$y_{p_2}'' + 2y_{p_2}' + y_{p_2} = xe^x \Leftrightarrow e^x(x+2) + 2ae^x(x+1) + ae^x x = xe^x$~~

~~$e^x(x+2+2x+2+x) = xe^x$~~

$y_{p_2}'' + 2y_{p_2}' + y_{p_2} = xe^x$

$e^x(ax + 2a + b) + 2e^x(ax + a + b) + e^x(ax + b) = xe^x$

$e^x(4ax + 4a + 4b) = xe^x$

$4a = 1 \Rightarrow a = \frac{1}{4}$

$4a + 4b = 0 \Rightarrow 1 + 4b = 0 \Rightarrow b = -\frac{1}{4}$

$y_{p_2} = e^x\left(\frac{1}{4}x - \frac{1}{4}\right)$

$y(x) = y_h(x) + y_{p_1}(x) + y_{p_2}(x) = \ldots$ sol. gen.

⑥ $y''' + y'' = \frac{x-1}{x^2}$

• y_h

$r^3 + r^2 = 0$

$r^2(r+1) = 0$

$r_1 = 0$, $m_{r_1} = 2$

$r_2 = -1$, $m_{r_2} = 1$

o bază pt. sp. sol. ec. omogene asociate este $\{f_1 = 1, f_2 = x, f_3 = e^{-x}\}$

$y_h = c_1 + c_2 x + c_3 e^{-x}$, $c_1, c_2, c_3 \in \mathbb{R}$

• y_p

$f(x) = \frac{x-1}{x^2}$ nu este cvasipolinom

folosim metoda variației constantelor

$y_p = c_1(x) + c_2(x) x + c_3(x) e^{-x}$

rezolvăm sistemul:

$$\begin{cases} c_1' + c_2' x + c_3' e^{-x} = 0 \\ c_2' - c_3' e^{-x} = 0 \\ c_3' e^{-x} = \frac{x-1}{x^2} \cdot \frac{1}{a_3} \end{cases}$$ coef. lui y''' (adică 1)

$c_3' e^{-x} = \frac{1}{x} - \frac{1}{x^2} \Rightarrow c_3' = \frac{e^x}{x} - \frac{e^x}{x^2}$ / $\int \Rightarrow c_3 = \frac{e^x}{x} + k_3$

$c_2' = c_3' e^x \Rightarrow \ln x + \frac{1}{x} + k_2 = c_2$

$c_1' = -c_2' x - c_3' e^x \Rightarrow c_1 = -x - \frac{1}{x} + k_1$

pt. $k_1 = k_2 = k_3 = 0$, avem $y_p(x) = (\ldots) = 1 - x + x\ln x$

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EC. DIF. EULER : $a_m(f(x))^m y^{(m)} + a_{m-1}(f(x))^{m-1} y^{(m-1)} + \dots + a_1 f(x) y' + a_0 y = g(x)$

1. Se identifică $f(x)$ și se face
   → sch. de fct. $z(t) = y(x)$ → se scoate $y(x)$ din $f(x) = e^t$
   → sch. de var. $e^t = f(x)$
2. se det. derivatele lui z
3. dăm în ec. dif. de ord. sup.

① $x^2y'' - 3xy' + 5y = 3x^2$, $x > 0$

sch. var. $x = e^t \Rightarrow t = \ln x$

sch. fct. $y(x) = y(e^t) = z(t)$

$\dot z(t) = e^t y'(e^t) = xy'(x) \Rightarrow xy'(x) = \dot z(t)$

$\ddot z(t) = e^t y'(e^t) + e^{2t} y''(e^t) = \dot z(t) + x^2 y''(x) \Rightarrow x^2 y''(x) = \ddot z(t) - \dot z(t)$

ec. devine:

$\ddot z(t) - \dot z(t) - 3\dot z(t) + 5z = 3e^{2t}$

$\ddot z(t) - 4\dot z(t) + 5z = 3e^{2t}$ ⊕ ec. dif. de ord. sup.

z_h

$r^2 - 4r + 5 = 0$

$\Delta = 16 - 20 = -4 \Rightarrow \sqrt{\Delta} = 2i$

$r_1 = \frac{4+2i}{2} = 2+i$

$r_2 = \frac{4-2i}{2} = 2-i$ $\Big\} \Rightarrow m_{r_1} = 1,\ m_{r_2} = 1$

$\{y_1 = e^{2t}\cos t,\ y_2 = e^{2t}\sin t\}$

$z_h = c_1 e^{2t}\cos t + c_2 e^{2t}\sin t$, $c_1, c_2 \in \mathbb{R}$

z_p

$f(x) = 3e^{2t} = e^{\alpha t}[P_m(t)\cos\beta t + Q_m(t)\sin\beta t]$

$\alpha = 2,\ \beta = 0 \Rightarrow \alpha + i\beta = 2$ nu este sol. a ec. caract. $\Rightarrow k = 0$

$P_m(x) = 3$

$Q_m(x) = 0$ $\Big\} \Rightarrow m = 0,\ m < -\infty \Rightarrow l < \max\{m, m\} = 0$

$z_p = x^k e^{\alpha t}[\tilde P_l(t)\cos\beta t + \tilde Q_l(t)\sin\beta t]$

$z_p = e^{2t} a$

$\dot z_p = e^{2t} a$

$\ddot z_p = e^{2t} 4a$

înlocuim în ec. ⊕

$4ae^{2t} + 2ae^{2t}(-4) + 5ae^{2t} = 3e^{2t}$

$ae^{2t} = 3e^{2t}$

$\boxed{a = 3}$

$z_p = 3e^{2t}$

$z(t) = z_h(t) + z_p(t) = e^{2t}(c_1\cos t + c_2\sin t + 3)$

$y(x) = x^2(c_1\cos(\ln x) + c_2\sin(\ln x) + 3)$

finală

㉘ p) $(x+1)^2 y'' - 3(x+1)y' + 4y = (x+1)^3$

sch. var. $x+1 = e^t \Rightarrow t = \ln(x+1)$

$\Rightarrow x = e^t - 1$

sch. fct. $z(t) = y(x) = y(e^t - 1)$

$\dot z(t) = e^t y'(e^t - 1) = (x+1)y'(x)$

$\ddot z(t) = e^t y'(e^t - 1) + e^t y''(e^t - 1)e^t = (x-1)y'(x) + (x-1)^2 y''(x) = \dots$

ec. devine:

$\dddot z(t) - \ddot z(t) - 3\dot z(t) + 4z(t) = e^{3t}$ ec. dif. ord. sup.

$\ddot z(t) - 4\dot z(t) + 4z(t) = e^{3t}$

z_h

$r^2 - 4r + 4 = 0$

$\Delta = 16 - 16 = 0$

$r_1 = r_2 = 2,\ m_r = 2 \Rightarrow \{y_1 = e^{2t},\ y_2 = te^{2t}\}$ bază c. p. sol. ec. dif. om.

$z_h(x) = c_1 e^{2t} + c_2 te^{2t}$, $c_1, c_2 \in \mathbb{R}$ (în loc. în x ...)

z_p

$f(x) = e^{3t} = e^{\alpha t}[P_m(t)\cos\beta t + Q_m(t)\sin\beta t]$

$\alpha = 3,\ \beta = 0 \Rightarrow \alpha + i\beta = 3$ nu este sol. ec. caract. $\Rightarrow k = 0$

$P_m(x) = 1 \Rightarrow m = 0$

$Q_m(x) = 0 \Rightarrow m < -\infty$ $\Big\} \Rightarrow l = 0$

$z_p = x^k e^{\alpha t}[\tilde P_l(t)\cos\beta t + \tilde Q_l(t)\sin\beta t]$

$z_p = e^{3t} a$

$\dot z_p = 3ae^{3t}$

$\ddot z_p = 9ae^{3t}$

înlocuim în ec.:

$(9a - 12a + 4a)e^{3t} = e^{3t}$

$\boxed{a = 1}$ $\Rightarrow z_p = e^{3t}$

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$z(t) = z_h(t) + z_p(t) = c_1 e^{2t} + c_2 t e^{2t} + e^{3t}$ $\Big\} \Rightarrow$

dar $t = \ln(x+1)$

$z(t) = y(x) = c_1 e^{2\ln(x+1)} + c_2 \ln(x+1) e^{2\ln(x+1)} + e^{3\ln(x+1)} =$

$= 2c_1 + 2\ln(x+1)c_2 + 3$

SISTEME DE EC. DIF. LIN.

DE ORD. I CU COEF. CONST.

OMOGENE

1. de la sist. scriem $\begin{pmatrix} x' \\ y' \\ z' \end{pmatrix} = A \begin{pmatrix} x \\ y \\ z \end{pmatrix}$ → mat. coef.
2. calculăm val. proprii λ prin $P_A(\lambda) = \det(A - \lambda I_3)$
3. pt. fiecare λ scriem $(A - \lambda I_3)\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$ și rezolvăm sist. omog. în funcție de o var. și $\begin{pmatrix} x \\ y \\ z \end{pmatrix} = y \begin{pmatrix} a \\ b \\ c \end{pmatrix}$ → vector ∀ var. din sist.
4. $Y(t) = e^{\lambda t} v, \ldots$ și $\{Y_1, Y_2 \ldots\}$ bază pt. sp. sol.
5. $\begin{pmatrix} x(t) \\ y(t) \\ z(t) \end{pmatrix} = c_1 Y_1(t) + c_2 Y_2(t) + \ldots$

① $\begin{cases} x' = 2x + y \\ y' = x + 3y - z \\ z' = 2y + 3z - x \end{cases}$, x, y, z funcții de t

scriem sist. ca:

$\begin{pmatrix} x' \\ y' \\ z' \end{pmatrix} = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 3 & -1 \\ -1 & 2 & 3 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix}$

$P_A(\lambda) = \det(A - \lambda I_3) = \begin{vmatrix} 2-\lambda & 1 & 0 \\ 1 & 3-\lambda & -1 \\ -1 & 2 & 3-\lambda \end{vmatrix} \overset{(\ldots)}{=} (2-\lambda)[(\lambda-3)^2 + 1]$

$\sigma(\lambda) = \{\lambda_1 = 2, \lambda_{2,3} = 3 \pm i\}$

$m_{\lambda_1} = 1$, $m_{\lambda_{2,3}} = 1 \Rightarrow$ A diagonalizează peste $\mathbb{C}$

$\lambda_1 = 2$

$A - 2I_3 = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 1 & -1 \\ -1 & 2 & 1 \end{pmatrix} \overset{(\ldots)}{\to} \begin{pmatrix} \textcircled{1} & 0 & -1 \\ 0 & \textcircled{1} & 0 \\ 0 & 0 & 0 \end{pmatrix}$

$(A - 2I_3)\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \Leftrightarrow \begin{cases} x_1 = x_3 \\ x_2 = 0 \end{cases} \Rightarrow x_1 = x_3$

$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = x_3 \underbrace{\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}}_{v_1} \Rightarrow \{v_1\}$ bază pt. $V_{\lambda_1=1} \Rightarrow \{e^{2t} v_1\}$ o sol. lim. indep. pt. sist. init.

$\lambda_2 = 3 + i$

$A - (3+i)I_3 = \begin{pmatrix} -1-i & 1 & 0 \\ 1 & -i & -1 \\ -1 & 2 & -i \end{pmatrix} \overset{(\ldots)}{\to} \begin{pmatrix} \textcircled{-1} & 2 & -i \\ 0 & \textcircled{2-i} & -i-1 \\ 0 & 0 & 0 \end{pmatrix}$

$(A - (3+i)I_3)\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \Leftrightarrow \begin{cases} x_1 = 2x_2 - i x_3 \\ (2-i)x_2 = (i+1)x_3 \end{cases} \overset{(\ldots)}{\Rightarrow} x_2 = \frac{1+3i}{5} x_3$, $x_1 = \frac{2+i}{5} x_3$

$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \frac{1}{5} x_3 \underbrace{\begin{pmatrix} 2+i \\ 1+3i \\ 5 \end{pmatrix}}_{w_1} \Rightarrow \{w_1\}$ bază pt. $V_{\lambda_2 = 3+i}$

$\Rightarrow \{\mathrm{Re}(e^{(3+i)t} w_1), \mathrm{Im}(e^{(3+i)t} w_2)\}$ SLi

② $\begin{cases} x' = y - 2x - 2z \\ y' = x - 2y + 2z \\ z' = 3x - 3y + 5z \end{cases}$ x, y, z funcții de t

$\begin{pmatrix} x' \\ y' \\ z' \end{pmatrix} = \begin{pmatrix} -2 & 1 & -2 \\ 1 & -2 & 2 \\ 3 & -3 & 5 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix}$

$P_A(\lambda) = \det(A - \lambda I_3) = \begin{vmatrix} -2-\lambda & 1 & -2 \\ 1 & -2-\lambda & 2 \\ 3 & -3 & 5-\lambda \end{vmatrix} \overset{C_1 = C_1 + C_2}{=} \begin{vmatrix} -1-\lambda & 1 & -2 \\ -1-\lambda & -2-\lambda & 2 \\ 0 & -3 & 5-\lambda \end{vmatrix} =$

$= (-1-\lambda)\begin{vmatrix} 1 & 1 & -2 \\ 1 & -2-\lambda & 2 \\ 0 & -3 & 5-\lambda \end{vmatrix} \overset{L_2 = L_2 - L_1}{=} -(1+\lambda)\begin{vmatrix} 1 & 1 & -2 \\ 0 & -3-\lambda & 4 \\ 0 & -3 & 5-\lambda \end{vmatrix} =$

$= -(1+\lambda)[-(3+\lambda)(5-\lambda) + 12] = -(1+\lambda)(\lambda-3)(\lambda+1)$

$\lambda_1 = -1$, $m_{\lambda_1} = 2$

$\lambda_2 = 3$, $m_{\lambda_2} = 1$

$\sigma(\lambda) = \{-1, 3\}$

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$\lambda_1 = -1$

$A + I_3 = \begin{pmatrix} -1 & 1 & -2 \\ 1 & -1 & 2 \\ 3 & -3 & 6 \end{pmatrix} \xrightarrow{(\dots)} \begin{pmatrix} -1 & 1 & -2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$

$(A + I_3)\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \Rightarrow x_2 = x_1 + 2x_3$

$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = x_1 \underbrace{\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}}_{v_1} + x_3 \underbrace{\begin{pmatrix} 0 \\ 2 \\ 1 \end{pmatrix}}_{v_2} \Rightarrow V_{\lambda_1=-1} = \left\{ \begin{pmatrix} x_1 \\ x_1 + 2x_3 \\ x_3 \end{pmatrix} \mid x_1, x_3 \in \mathbb{R} \right\}$

$\Rightarrow \{v_1, v_2\}$ o bază pt. $V_{\lambda_1=-1}$

$\lambda_2 = 3$

$A - 3I_3 = \begin{pmatrix} & & \\ & & \\ & & \end{pmatrix} \dots$

$\Rightarrow \{v_3\}$ o bază pt. $V_{\lambda_2=3}$, unde $v_3 \in \begin{pmatrix} -1 \\ 1 \\ 3 \end{pmatrix}$

$\lambda_1 = -1 \in \mathbb{R} \Rightarrow Y_1(t) = c_1 e^{-t}\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$, $Y_2(t) = c_2 e^{-t}\begin{pmatrix} 0 \\ 2 \\ 1 \end{pmatrix}$

$\lambda_2 = 3 \in \mathbb{R} \Rightarrow Y_3(t) = c_3 e^{3t}\begin{pmatrix} -1 \\ 1 \\ 3 \end{pmatrix}$

$\Rightarrow \begin{pmatrix} x(t) \\ y(t) \\ z(t) \end{pmatrix} = c_1 e^{-t}\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} + c_2 e^{-t}\begin{pmatrix} 0 \\ 2 \\ 1 \end{pmatrix} + c_3 e^{3t}\begin{pmatrix} -1 \\ 1 \\ 3 \end{pmatrix}$

(28) a) $\begin{cases} x' = 3x - y + z \\ y' = -x + 5y - z \\ z' = x - y + 3z \end{cases}$, $x(0) = 1,\ y(0) = 0,\ z(0) = 1$

$= A$

$\begin{pmatrix} x' \\ y' \\ z' \end{pmatrix} = \begin{pmatrix} 3 & -1 & 1 \\ -1 & 5 & -1 \\ 1 & -1 & 3 \end{pmatrix}\begin{pmatrix} x \\ y \\ z \end{pmatrix}$

$P_A(\lambda) = \det(A - \lambda I_3) = \begin{vmatrix} 3-\lambda & -1 & 1 \\ -1 & 5-\lambda & -1 \\ 1 & -1 & 3-\lambda \end{vmatrix} \overset{C_1 = C_1 + C_3}{=\!=\!=} \begin{vmatrix} 4-\lambda & -1 & 1 \\ -2 & 5-\lambda & -1 \\ 4-\lambda & -1 & 3-\lambda \end{vmatrix} =$

$\overset{C_1 = C_1 + C_2}{=\!=\!=} \begin{vmatrix} 3-\lambda & -1 & 1 \\ 3-\lambda & 5-\lambda & -1 \\ 3-\lambda & -1 & 3-\lambda \end{vmatrix} = (3-\lambda)\begin{vmatrix} 1 & -1 & 1 \\ 1 & 5-\lambda & -1 \\ 1 & -1 & 3-\lambda \end{vmatrix} \overset{\substack{L_2 = L_2 - L_1 \\ L_3 = L_3 - L_1}}{=\!=\!=}$

$= \begin{vmatrix} 1 & -1 & 1 \\ 0 & 6-\lambda & -2 \\ 0 & 0 & 2-\lambda \end{vmatrix} = (3-\lambda)(6-\lambda)(2-\lambda)$

$\sigma(A) = \{3, 6, 2\}$

$m_{\lambda_1} = m_{\lambda_2} = m_{\lambda_3} = 1 \Rightarrow A$ diagonalizează

• $\lambda_1 = 2$

$A - 2I_3 = \begin{pmatrix} 1 & -1 & 1 \\ -1 & 3 & -1 \\ 1 & -1 & 1 \end{pmatrix} \xrightarrow[L_3 = L_3 - L_1]{L_2 = L_2 + L_1} \begin{pmatrix} \textcircled{1} & -1 & 1 \\ 0 & \textcircled{2} & 0 \\ 0 & 0 & 0 \end{pmatrix}$

$(A - 2I_3)\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \Leftrightarrow \begin{cases} x_1 = x_2 - x_3 \Rightarrow x_1 = -x_3 \\ x_2 = 0 \end{cases}$

$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = x_3 \underbrace{\begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}}_{v_1} \Rightarrow V_{\lambda_1=2} = \left\{ \begin{pmatrix} -x_3 \\ 0 \\ x_3 \end{pmatrix} \mid x_1 \in \mathbb{R} \right\}$

$\Rightarrow \{v_1\}$ o bază pt. $V_{\lambda_1=2}$

mat. $Y_1(t) = e^{\lambda t} v = e^{2t}\begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}$

• $\lambda_2 = 3$

$A - 3I_3 = \begin{pmatrix} 0 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 0 \end{pmatrix} \rightarrow \begin{pmatrix} -1 & 2 & -1 \\ 0 & -1 & 1 \\ 1 & -1 & 0 \end{pmatrix} \xrightarrow{L_3 = L_3 + L_1} \begin{pmatrix} -1 & 2 & -1 \\ 0 & -1 & 1 \\ 0 & 1 & -1 \end{pmatrix} \xrightarrow{L_3 = L_3 + L_2}$

$\rightarrow \begin{pmatrix} \textcircled{-1} & 2 & -1 \\ 0 & \textcircled{-1} & 1 \\ 0 & 0 & 0 \end{pmatrix}$

$(A - 3I_3)\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \Leftrightarrow \begin{cases} 2x_2 = x_1 + x_3 \Rightarrow 2x_3 = x_1 + x_3 \Rightarrow x_1 = x_3 \\ x_2 = x_3 \end{cases}$

$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = x_3 \underbrace{\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}}_{v_2} \Rightarrow V_{\lambda_2=3} = \left\{ \begin{pmatrix} x_3 \\ x_3 \\ x_3 \end{pmatrix} \mid x_3 \in \mathbb{R} \right\}$

$\Rightarrow \{v_2\}$ o bază pt. $V_{\lambda_2=3}$

mat. $Y_2(t) = e^{3t}\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$

• $\lambda_3 = 6$

$A - 6I_3 = \begin{pmatrix} -3 & -1 & 1 \\ -1 & -1 & -1 \\ 1 & -1 & -3 \end{pmatrix} \rightarrow \begin{pmatrix} -1 & -1 & -1 \\ -3 & -1 & 1 \\ 1 & -1 & -3 \end{pmatrix} \xrightarrow[L_3 = L_3 + L_1]{L_2 = L_2 - 3L_1}$

$\rightarrow \begin{pmatrix} -1 & -1 & -1 \\ 0 & 2 & 4 \\ 0 & -2 & -4 \end{pmatrix} \xrightarrow{L_3 = L_3 + L_2} \begin{pmatrix} \textcircled{-1} & -1 & -1 \\ 0 & \textcircled{2} & 4 \\ 0 & 0 & 0 \end{pmatrix}$

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$(A-6I_3)\begin{pmatrix}x_1\\x_2\\x_3\end{pmatrix}=\begin{pmatrix}0\\0\\0\end{pmatrix} \Rightarrow \begin{cases}-x_1-x_2-x_3=0 \Rightarrow -x_1+2x_3-x_3=0 \Rightarrow x_1=x_3\\ 2x_2+4x_3=0 \Rightarrow x_2=-2x_3\end{cases}$

$\begin{pmatrix}x_1\\x_2\\x_3\end{pmatrix}=x_3\underbrace{\begin{pmatrix}1\\-2\\1\end{pmatrix}}_{v_3} \Rightarrow V_{\lambda_3=6}=\left\{\begin{pmatrix}x_3\\-2x_3\\x_3\end{pmatrix} \mid x_3\in\mathbb{R}\right\}$

$\Rightarrow \{v_3\}$ o baza pt. $V_{\lambda_3=6}$

mat. $Y_3=e^{6t}\begin{pmatrix}1\\-2\\1\end{pmatrix}$

$\begin{pmatrix}x(t)\\y(t)\\z(t)\end{pmatrix}=c_1Y_1(t)+c_2Y_2(t)+c_3Y_3(t)$

$\begin{pmatrix}x(t)\\y(t)\\z(t)\end{pmatrix}=c_1e^{2t}\begin{pmatrix}-1\\0\\1\end{pmatrix}+c_2e^{3t}\begin{pmatrix}1\\1\\1\end{pmatrix}+c_3e^{6t}\begin{pmatrix}1\\-2\\1\end{pmatrix}$

$\begin{pmatrix}1\\0\\1\end{pmatrix}=c_1\begin{pmatrix}-1\\0\\1\end{pmatrix}+c_2\begin{pmatrix}1\\1\\1\end{pmatrix}+c_3\begin{pmatrix}1\\-2\\1\end{pmatrix}$

det. sist.:

$\begin{cases}-c_1+c_2+c_3=1\\ c_2-2c_3=0\\ c_1+c_2+c_3=1\end{cases}$

$(1)+(3) \Rightarrow 2c_2+2c_3=2 \Rightarrow \begin{cases}c_2=1-c_3\\ c_2=+2c_3\end{cases} \Rightarrow 1-c_3=2c_3 \Rightarrow \boxed{c_3=\frac{1}{3}}$

$\Rightarrow \boxed{c_2=\frac{2}{3}} \Rightarrow \boxed{c_1=0}$

$\begin{cases}x(t)=-c_1e^{2t}+c_2e^{3t}+c_3e^{6t}=\frac{2}{3}e^{3t}-\frac{1}{3}e^{6t}\\ y(t)=0c_1e^{2t}+c_2e^{3t}-2c_3e^{6t}=\frac{2}{3}e^{3t}-\frac{2}{3}e^{6t}\\ z(t)=c_1e^{2t}+c_2e^{3t}+c_3e^{6t}=\frac{2}{3}e^{3t}+\frac{1}{3}e^{6t}\end{cases}$

NEOMOGENE

① $\begin{cases}x_1'=x_2-\sin t\\ x_2'=2x_1+x_2\end{cases}, t\in\mathbb{R}$

sol. gen. $\begin{pmatrix}x_1\\x_2\end{pmatrix}=\begin{pmatrix}x_1\\x_2\end{pmatrix}_p+\begin{pmatrix}x_1\\x_2\end{pmatrix}_h$

rescriem: $\begin{pmatrix}x_1'\\x_2'\end{pmatrix}=\underbrace{\begin{pmatrix}0&1\\2&1\end{pmatrix}}_{A}\begin{pmatrix}x_1\\x_2\end{pmatrix}+\underbrace{\begin{pmatrix}-\sin t\\0\end{pmatrix}}_{b(t)}$

calc. val. proprii:

$P_A(\lambda)=\det(A-\lambda I_2)=\begin{vmatrix}-\lambda&1\\2&1-\lambda\end{vmatrix}=\lambda(1-\lambda)-2=\lambda^2-\lambda-2=$

$=(\lambda-2)(\lambda+1)$

$\sigma(A)=\{-1,2\}$

$m_{\lambda_1}=m_{\lambda_2}=1 \Rightarrow$ A diagonalizabila

- $\lambda_1=-1$

$A+I_2=\begin{pmatrix}1&1\\2&2\end{pmatrix}\to\begin{pmatrix}\textcircled{1}&1\\0&0\end{pmatrix}$

$(A+I_2)\begin{pmatrix}a\\b\end{pmatrix}=\begin{pmatrix}0\\0\end{pmatrix} \Rightarrow a+b=0 \Rightarrow a=-b$

$\begin{pmatrix}a\\b\end{pmatrix}=a\underbrace{\begin{pmatrix}1\\-1\end{pmatrix}}_{v_1} \Rightarrow V_{\lambda_1=-1}=\left\{\begin{pmatrix}a\\-a\end{pmatrix} \mid a\in\mathbb{R}\right\}=Sp\left\{\begin{pmatrix}1\\-1\end{pmatrix}\right\}$

$\Rightarrow \{v_1\}$ o baza pt. $V_{\lambda_1=-1}$

mat. $Y_1=e^{\lambda t}v=e^{-t}\begin{pmatrix}1\\-1\end{pmatrix}$

- $\lambda_2=2$

$A-2I_2=\begin{pmatrix}-2&1\\2&-1\end{pmatrix}\to\begin{pmatrix}\textcircled{-2}&1\\0&0\end{pmatrix}$

$(A-2I_2)\begin{pmatrix}a\\b\end{pmatrix}=\begin{pmatrix}0\\0\end{pmatrix} \Rightarrow -2a+b=0 \Rightarrow b=2a$

$\begin{pmatrix}a\\b\end{pmatrix}=a\underbrace{\begin{pmatrix}1\\2\end{pmatrix}}_{v_2} \Rightarrow V_{\lambda_2=2}=\left\{\begin{pmatrix}a\\2a\end{pmatrix} \mid a\in\mathbb{R}\right\}=Sp\left\{\begin{pmatrix}1\\2\end{pmatrix}\right\}$

$\Rightarrow \{v_2\}$ o baza pt. $V_{\lambda_2=2}$

mat. $Y_2=e^{\lambda t}v=e^{2t}\begin{pmatrix}1\\2\end{pmatrix}$

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deci, $\{\varphi_1, \varphi_2\}$ este o bază pt. sp. sol. al. sist.

$x_h(t) = c_1\varphi_1(t) + c_2\varphi_2(t)$

$\begin{pmatrix} x_1(t) \\ x_2(t) \end{pmatrix}_h = c_1 e^{-t}\begin{pmatrix} 1 \\ -1 \end{pmatrix} + c_2 e^{2t}\begin{pmatrix} 1 \\ 2 \end{pmatrix}$

căutăm $x_p(t)$ prin **METODA VARIAȚIEI CONSTANTELOR**

$x_p(t) = c_1(t)\varphi_1(t) + c_2(t)\varphi_2(t)$

unde c_1', c_2' verifică $c_1'\varphi_1(t) + c_2'\varphi_2(t) = b(t)$

$c_1' e^{-t}\begin{pmatrix} 1 \\ -1 \end{pmatrix} + c_2' e^{2t}\begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} -5\cos t \\ 0 \end{pmatrix}$

$\begin{cases} c_1' e^{-t} + c_2' e^{2t} = -5\cos t \\ -c_1' e^{-t} + 2c_2' e^{2t} = 0 \end{cases}$ ⊕

$3c_2' e^{2t} = -5\cos t$

$c_2' = -\frac{5}{3}\cdot\frac{\cos t}{e^{2t}} \quad | \int$

$c_1 = -\frac{5}{3}\int \frac{\cos t}{e^{2t}}\,dt = -\frac{e^{-2t}}{3}(\sin t - 2\cos t) + a_1$

$\overset{(\dots)}{\Rightarrow} c_1 = \frac{5}{3}e^{t}(\cos t + \sin t) + a_2$

pt. $a_1 = a_2 = 0$

$x_p(t) = -\frac{e^{-2t}}{3}(\sin t - 2\cos t)e^{2t}\begin{pmatrix} 1 \\ 2 \end{pmatrix} + \frac{5}{3}e^{t}(\cos t + \sin t)e^{-t}\begin{pmatrix} 1 \\ -1 \end{pmatrix}$

$= \begin{pmatrix} -\cos t - 2\sin t \\ 3\cos t + \sin t \end{pmatrix}$

(28) A) $\begin{cases} x' + 2x + y = \sin t \\ y' - 4x - 2y = \cos t \end{cases}$

$\begin{pmatrix} x' \\ y' \end{pmatrix} = \underbrace{\begin{pmatrix} -2 & -1 \\ 4 & 2 \end{pmatrix}}_{A}\begin{pmatrix} x \\ y \end{pmatrix} + \underbrace{\begin{pmatrix} \sin t \\ \cos t \end{pmatrix}}_{b(t)}$

$\begin{pmatrix} x \\ y \end{pmatrix}_h$: rezolvăm sistemul omogen asociat

$P_A(\lambda) = \det(A - \lambda I_2) = \begin{vmatrix} -2-\lambda & -1 \\ 4 & 2-\lambda \end{vmatrix} = -(2+\lambda)(2-\lambda) + 4 =$

$= -(4-\lambda^2) + 4 = -4 + \lambda^2 + 4$

$\sigma(A) = \{0\}$

$m_\lambda = 2$

$A - \lambda I_2 = A = \begin{pmatrix} -2 & -1 \\ 4 & 2 \end{pmatrix} \to \begin{pmatrix} \textcircled{2} & -1 \\ 0 & 0 \end{pmatrix}$ $m_{g\lambda}$ = nr. col. FĂRĂ pivot !

A val. pr. simplă ⇒ diag.

$\dim_{\mathbb{R}} V_\lambda = 2 - 1 = 1 \Rightarrow m_{g\lambda} = 1$

$\left.\begin{matrix} \\ \text{dar } m_{a\lambda} = 2 \end{matrix}\right\} \Rightarrow$ A nu diagonalizează